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The Evolution of Cauchy's Closed Curve Theorem and Newman's Simple Proof

Joseph Bak and Strashimir G. Popvassilev

Abstract. We examine the development of Cauchy's closed curve theorem, including the early contributions of Clairaut, d'Alembert, Cauchy himself, Goursat, and Pringsheim, as well as more recent approaches due to Ahlfors, Rudin, and others. A particularly simple proof was given by D. J. Newman, utilizing his original definition of a simply-connected region in the (complex) plane. We show that this definition is equivalent to the other, more familiar definitions of simple-connectedness so that Newman's approach offers an alternative and very elegant proof of the general result.

1. INTRODUCTION. On August 22, 1814, one day after his 25th birthday, Cauchy presented a long memoir on definite integrals to Classe I of the Institut de France. The memoir was very quickly examined and recommended for publication by Legendre and Lacroix. However, because of the changing political situation (Napoleon abdicated in 1814 and again in 1815), the intended journal ceased publication and the memoir [**6**] wasn't published until 1827 [**19**, p. 24]. By that time, Cauchy's subsequent 1825 memoir [**7**] extended his analysis of definite integrals to include complex limits. The 1825 memoir also contained the genesis of one of the most pivotal results in the development of complex analysis, what has come to be known as Cauchy's closed curve theorem. Also referred to as Cauchy's integral theorem, it asserts that if f' exists at all points of a simply-connected region R, then $\int_C f(z)\,dz = 0$ for any piecewise-smooth closed curve C contained in R. (Note that a "region" denotes an open, connected set. A simply-connected region can be loosely defined as one "without holes." The formal definition is given in Section 7, below.)

This theorem opens the door to a wide range of results, including the remarkable fact that if f' exists throughout an open set S, then f is infinitely differentiable in S, and, in fact, f has a power series representation in a neighborhood of any point of S. For that reason, f is said to be "analytic" when f' exists throughout an open set S.

The simple-connectedness of the region is not only sufficient but also necessary. As we will see below, if $\int_C f(z)\,dz = 0$ for all functions analytic in a region R and for all piecewise-smooth closed curves C contained in R, then R must be simply-connected. There is also a partial converse regarding the differentiability of f. That is, if $\int_C f(z)\,dz = 0$ for all piecewise-smooth curves in an open set R and if f is continuous there, then f is also differentiable in R.

The conclusion of the theorem can be extended to include all rectifiable curves, but that is a fairly minor extension. Most authors, beginning with Cauchy, have preferred to deal with piecewise-smooth curves since, in that case, if C is given by $\phi(t)$, $0 \le t \le 1$, then $\int_C f(z)\,dz = \int_0^1 f(\phi(t))\phi'(t)\,dt$. If C is rectifiable but not necessarily piecewise-smooth, $\int_C f(z)\,dz$ may be defined as the limit of a Riemann sum. (An alternative approach is outlined in Section 7, below.) In any case, for piecewise-smooth curves,

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the definitions are equivalent, and once the theorem is proved for piecewise-smooth curves, an easy argument shows that it applies as well to all rectifiable curves.

The theorem is justifiably known as Cauchy's theorem for his original proof (albeit under a somewhat stronger hypotheses and with a less general conclusion) and for his elaboration on the theorem in his development of complex analysis. In the century that followed Cauchy's original proof, the mathematical world acquired the formal "ε - δ" approach to limits, continuity, and differentiability, the Jordan curve theorem, and the definition of a simply-connected set. Obviously, all of these ideas had an impact on the final form of Cauchy's theorem. Remarkably, though, Cauchy's original ideas, as well as those of Goursat, already provided the core of the proof of the general result.

Considering the importance of the closed curve theorem, one might expect that some reference to it would be found among Gauss's many contributions to complex analysis. In fact, in a letter that Gauss wrote to his friend Friedrich Wilhelm Bessel in 1811 but which was not published until 1880, Gauss wrote about the theorem: "This is a very beautiful theorem, for which I will give a not difficult proof at a suitable opportunity." [**4**, p. 8] Unfortunately, much like Fermat's famous marginal note, whatever proof Gauss had in mind, never actually appeared.

In what follows, we explore the evolution of the proof, including some variations that involve nonstandard definitions of simple-connectedness. And we close with an argument showing that D. J. Newman's original approach provides an extremely straightforward proof of the general result.

2. BACKGROUND. The theorem can be traced back to 1740 when Clairaut pointed out that if a differential form $P(x, y)\,dx + Q(x, y)\,dy$ is exact; i.e., if $P = F_x$ and $Q = F_y$ for some C^1 function $F(x, y)$, then

$$P_y = Q_x. \tag{1}$$

Moreover, he showed that if a differential form is exact, then $\int_C P\,dx + Q\,dy$ is independent of the path of the curve and dependent only on the integrand and the initial and terminal points of the curve. It follows that for exact forms $\int_C P\,dx + Q\,dy = 0$ when C is a closed curve. Finally (and incorrectly), he went on to say that $P(x, y)\,dx + Q(x, y)\,dy$ would be exact if and only if $P_y = Q_x$ [**8**, pp. 294–297]. In 1768, the error was pointed out by d'Alembert, citing the counterexample that has since been repeated in hundreds of textbooks on vector calculus. That is,

$$\int_{C(0,r)} \frac{y\,dx - x\,dy}{x^2 + y^2} = -2\pi$$

if $C(0, r)$ is any circle centered at the origin and surrounding it in a counterclockwise manner, in spite of the fact that a direct calculation shows $P_y = Q_x$ if

$$P(x, y) = \frac{y}{x^2 + y^2}\,; \quad Q(x, y) = \frac{-x}{x^2 + y^2}\,.$$

All of the above relates directly to integrals of complex-valued functions in the complex plane since $\int_C f(z)\,dz = \int_C [u(x, y) + i\,v(x, y)](dx + i\,dy) = \int_C (u\,dx - v\,dy) + i\int_C (v\,dx + u\,dy)$, where u, v are the real and imaginary parts of f, respectively. Moreover, the functions u and v satisfy what were to become known as the Cauchy–Riemann equations: $u_x = v_y$; $u_y = -v_x$ so that the differential forms in both the real and imaginary parts of $\int_C f(z)\,dz$ satisfy condition (1). (In fact, the Cauchy–Riemann equations appeared in an essay by d'Alembert in 1752 well before Cauchy

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(1789–1857) or Riemann (1826–1866) was born.) So d'Alembert's counterexample to Clairaut's incorrect characterization of exact differential forms also offers a counterexample in complex analysis. It shows that even if f is analytic along a closed curve C, $\int_C f(z)\,dz$ may be nonzero since

$$\int_{C(0,r)} \frac{1}{z}\,dz = \int_{C(0,r)} \frac{x\,dx + y\,dy}{x^2+y^2} + i\int_{C(0,r)} \frac{-y\,dx + x\,dy}{x^2+y^2} = 2\pi i.$$

3. CAUCHY'S CONTRIBUTION TO CAUCHY'S THEOREM. Cauchy's original version of the closed curve theorem obviously took d'Alembert's counterexample into consideration. Note that proving the integral around any closed curve in a region is zero is equivalent to showing that the integrals along two different paths in the region with the same endpoints are the same. What Cauchy proved, cast in current terminology, is the following.

Theorem. *If f is continuously differentiable (with respect to z) in a convex region R and if α and β are piecewise-smooth curves with the same endpoints, then $\int_\alpha f(z)\,dz = \int_\beta f(z)\,dz$.*

Proof. Suppose $\alpha(t)$ and $\beta(t)$, $0 \le t \le 1$, are two smooth curves in R with $\alpha(0) = \beta(0)$ and $\alpha(1) = \beta(1)$. Let $\gamma(t) = \beta(t) - \alpha(t)$ and let C_s be the curve defined by

$$C_s(t) = \alpha(t) + s\,\gamma(t) = (1-s)\alpha(t) + s\,\beta(t),\ 0 \le s, t \le 1.$$

(Since R is convex, C_s is contained in R for $0 \le s \le 1$.) Let

$$I(s) = \int_{C_s} f(z)\,dz = \int_0^1 f(z(t))\frac{dz}{dt}\,dt,$$

where $z(t) = C_s(t)$.

Then, $I(s+h) - I(s)$

$$= \int_0^1 f(z+\Delta z)[\alpha'(t) + (s+h)\gamma'(t)]\,dt - \int_0^1 f(z)[\alpha'(t) + s\gamma'(t)]\,dt$$

$$= \int_0^1 [f(z+\Delta z) - f(z)][\alpha'(t) + s\gamma'(t)]\,dt + \int_0^1 f(z+\Delta z)h\gamma'(t)\,dt,$$

with $\Delta z = h\gamma(t)$. Expanding $f(z+\Delta z)$, we find that the coefficient of h in $I(s+h) - I(s)$ is given by

$$\int_0^1 \gamma(t) f'(z(t))z'(t)\,dt + \int_0^1 f(z(t))\gamma'(t)\,dt$$

$$= \int_0^1 [f(z(t))\gamma(t)]'\,dt = f(z(t))\gamma(t)\big|_0^1 = 0,$$

since $\gamma(0) = \gamma(1) = 0$. It follows that $I(s+h) - I(s) = o(h)$. Hence, $I'(s) = 0$ and $\int_\alpha f(z)\,dz = I(0) = I(1) = \int_\beta f(z)\,dz$. ∎

For historical accuracy, it should be noted that Cauchy's memoir [**7**] (from 1825) reflected the style of its time, rather than the familiar type of mathematical rigor that

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was introduced later in the 19th century. For example, his stated condition on f was that it should be "finite and continuous." Since he went on to discuss the integral of f', he clearly assumed that f was differentiable. In fact, Cauchy's idea of a continuous function (at this time) was some type of analytic expression. The seemingly unnecessary adjective "finite" meant that it had no poles. Cauchy also assumed that $I(s)$ was a differentiable function of s and that $f(z + \Delta z)$ could be expanded in powers of Δz. And although Cauchy's proof applied to any convex region, he actually limited his result to a rectangle. He assumed that f was analytic in the rectangle R with opposite vertices at two complex values z_0 and z_1 and that all the paths considered were "monotonic" from z_0 to z_1. That is, for curves defined by $x(t) + i\, y(t)$, he assumed that both $x(t)$ and $y(t)$ were monotonic (real-valued) functions of t, for $0 \le t \le 1$. With those assumptions, all of the curves remained within the rectangle R.

Although some of the mathematical assumptions of its time may appear naïve today, Cauchy's proof was nevertheless quite remarkable. His introduction of $C_s(t)$ to evolve from $\alpha(t) = C_0(t)$ to $\beta(t) = C_1(t)$ not only established the result for convex regions, but it also anticipated the general proof for any simply-connected region. (See Sections 6 and 7 below.) But he needed a little help to complete the general result.

4. FROM CAUCHY TO GOURSAT. Fifty years after Cauchy's 1825 memoir, Briot and Bouquet [**5**] offered a somewhat neater version of Cauchy's argument to prove that the integral of f around a piecewise-smooth closed curve C is zero. They assumed that C lies within a region R which is "star-shaped" with respect to the origin, i.e., a region that contains all points of the form tz, $0 \le t \le 1$, for every $z \in R$). Let

$$I(s) = \int_0^1 f(s\phi(t))s\phi'(t)\,dt,$$

where C is given by $\phi(t)$, $0 \le t \le 1$, and $0 \le s \le 1$. In an argument similar to Cauchy's, they showed that $I'(s) = 0$ so that $\int_C f(z)\,dz = I(1) = I(0) = 0$. Unlike Cauchy, however, they sought to avoid any additional assumptions about the analytic function f, relying instead only on the definition of differentiability. Their argument, which contained a small but critical flaw, was as follows. Note that $I(s)$ is clearly continuous for $0 \le s \le 1$ and, for $0 < s < 1$,

$$I(s + \Delta s) - I(s) = \int_0^1 f(z + \Delta z)(s + \Delta s)\phi'(t)\,dt - \int_0^1 f(z)s\phi'(t)\,dt,$$

where $z = s\phi(t)$ and $\Delta z = \Delta s \cdot \phi(t)$. The above equality extends as

$$= \int_0^1 [f(z + \Delta z) - f(z)]s\phi'(t)\,dt + \int_0^1 f(z + \Delta z)\Delta s \cdot \phi'(t)\,dt.$$

By the definition of differentiability, $f(z + \Delta z) - f(z) = f'(z)\Delta z + \varepsilon_z \Delta z$ where $\varepsilon_z \to 0$ as $\Delta z \to 0$, so that

$$\frac{I(s + \Delta s) - I(s)}{\Delta s} = \int_0^1 f'(z)\phi(t)s\phi'(t)\,dt + \int_0^1 \varepsilon_z \phi(t)s\phi'(t)\,dt + \int_0^1 f(z + \Delta z)\phi'(t)\,dt$$

and, as $\Delta s \to 0$, the middle integral approaches 0. Hence,

$$I'(s) = \int_0^1 [zf'(z) + f(z)]\phi'(t)\,dt = \frac{1}{s}\int_{C_s} [zf'(z) + f(z)]\,dz = \frac{1}{s}\int_{C_s} G'(z)\,dz,$$

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where C_s denotes the curve defined by $C_s(t) = s\phi(t)$ and $G(z) = zf(z)$. Finally, since C_s is a closed curve, the latter integral, and hence $I'(s)$, are both zero [**5**, pp. 117–121]. ■

The flaw in the above proof is the following. As noted above, the differentiability of f at z guarantees that $f(z + \Delta z) - f(z) = \Delta z \cdot f'(z) + \varepsilon_z \cdot \Delta z$ where $\varepsilon_z \to 0$ as $\Delta z \to 0$. So for any *fixed point* z and any given $\varepsilon > 0$, there exists $\delta > 0$ such that $|\Delta z| < \delta$ guarantees $|\varepsilon_z| < \varepsilon$. However, to prove that $\int_0^1 \varepsilon_z \phi(t) s\phi'(t)\, dt \to 0$ as $\Delta s \to 0$, it must be established that for some sufficiently small Δs (and correspondingly small Δz), $|\varepsilon_z| < \varepsilon$ for *all points* z on the curve C_s. Whether this requires something more than just the differentiability of f became a subplot in the continuing story of the proof as further developed by Goursat.

5. GOURSAT, WITH A LITTLE HELP FROM PRINGSHEIM. Like Briot and Bouquet, Goursat offered a proof that focused primarily on the differentiability of f. His proof, however, was more direct. As he put it: "Here is a proof of Cauchy's theorem which seems to me a little simpler than the usual proofs. It relies only on the definition of the derivative and on the remark that the definite integrals $\int dz$ *and* $\int z\, dz$, taken along any closed contour, are equal to zero." [**11**] Despite this modest introduction, Goursat actually sought to prove that $\int_C f(z)\, dz = 0$ along *any* rectifiable closed curve C, as long as f is differentiable at all points inside and on the curve. He argued as follows.

Suppose the interior of C is partitioned by a grid of horizontal and vertical lines with a (small) distance l between adjacent parallel lines. Then the integral along C is equivalent to the sum of the integrals along (the boundaries of) the various subsets into which C and its "inside" have been partitioned. These subsets will be of two types: a complete square or a truncated square with a portion of its perimeter replaced by a part of C. In both cases, the maximum distance between any two points in the subregion will be at most $l\sqrt{2}$. Let z_i be a point in subregion R_i with boundary S_i. Then $f(z) = f(z_i) + (z - z_i) f'(z_i) + \varepsilon_i (z - z_i)$. Since the integral of any linear function around a closed curve is zero,

$$\int_{S_i} f(z)\, dz = \int_{S_i} \varepsilon_i (z - z_i)\, dz. \tag{2}$$

Now, for any $\varepsilon > 0$, let l be small enough to guarantee that $|\varepsilon_i| < \varepsilon$. If R_i is a square, (2) is bounded in modulus by $4\sqrt{2}l^2\varepsilon = 4\sqrt{2}a_i\varepsilon$, where a_i denotes the area of the square R_i. And the sum over all such regions is bounded by $4\sqrt{2}A\varepsilon$, where A is the area of all the complete squares inside C. If R_i is a region whose boundary includes a subset C_i of C, (2) is bounded by $\sqrt{2}l\varepsilon(4l + length(C_i)) < \varepsilon\sqrt{2}(4a_i + length(C_i))$. Summing over all of these subregions and combining with the sum over all squares inside C shows that $\int_C f(z)\, dz$ is bounded in modulus by $\sqrt{2}(4B + L)\varepsilon$, where B is the sum of areas of all the squares partially or totally inside C, and L is the length of C. Hence, $\int_C f(z)\, dz = 0$. ■

Goursat addressed the question of finding a value of l sufficiently small so that $|\varepsilon_i| < \varepsilon$ for all i. In his introduction, he says: "Let $f(z)$ be a function of the complex variable z, uniform and continuous [i.e., uniformly continuous?], as well as its first derivative $f'(z)$, in the interior of a region bounded by a simple or multiple closed curve C, and on the curve itself." Of course, Goursat need not have mentioned the uniform continuity since that follows automatically for any function that is continuous on a compact set. For that matter, the continuity of f follows from the differentiability

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of f. But, in any case, he did include the assumption that f' was continuous. Then, at the end of his paper, he wrote: "In truth, the proof assumes, in order to be absolutely rigorous, that one can choose the length l sufficiently small so that the moduli of all quantities designated by ε [that is, ε_i] should be all less than a number chosen in advance, as small as one would like. This results from the hypothesis made about $f(z)$ and its derivative."

Goursat's 1884 article seems to have attracted a good deal of attention, some of it, unfortunately, negative. In particular, Pringsheim argued that the article really offered nothing new, and among other comments, he questioned whether continuity of the derivative is really necessary [**10**]. The question was answered subsequently by Goursat, who wrote in a second article [**12**]: "I have recognized for a long time that the proof of Cauchy's Theorem, which I gave in 1883 [published in 1884], need not assume the continuity of the derivative. In order to respond to the request expressed to me by Professor Wm. F. Osgood, I will quickly show here how one can make this extension." His argument, which has come to be known as Goursat's lemma, was the following.

Lemma. *Let $f(z)$ be differentiable at all points of a compact set R consisting of a closed contour C and its "inside," and let ε be an arbitrary positive number. Then one can decompose R into sufficiently small subsets such that in each subset R_i there exists a point z_i with*

$$|f(z) - f(z_i) - (z - z_i)f'(z_i)| \leq \varepsilon|z - z_i| \tag{3}$$

throughout the boundary of R_i.

The proof of the lemma is by a form of continued bisection. Suppose R is partitioned by a grid of equally spaced horizontal and vertical lines, with the distance between adjacent parallel lines equal to λ. If (3) is not true, then there exists at least one subset $R^{(1)}$ in which (3) does not hold. But then $R^{(1)}$ can be further subdivided by a grid of lines with distance $\lambda/2$ between adjacent parallel lines, and if (3) does not hold in $R^{(1)}$, it must fail to be true in at least one of the new subsets, which we can denote by $R^{(2)}$. In this manner, we can form a nested sequence of subsets $R^{(1)} \supset R^{(2)} \supset R^{(3)} \cdots$ such that (3) fails to hold in each one, and such that $\cap R^{(i)}$ is equal to a single point z_0. But then, for sufficiently large n, $R^{(n)}$ is contained in any disk of radius $\delta > 0$ centered at z_0 , and since f is differentiable at z_0, the contradiction is obvious. ■

While this aspect of Goursat's proof was settled, there remained the issue of whether the proof was actually valid for any rectifiable curve. Goursat's 1884 article was followed several years later by the first proof of the Jordan curve theorem. This theorem justified Goursat's reference to the inside of a curve (what he referred to simply as the area bounded by the curve), at least for simple closed curves. On the other hand, the theorem also raised an awareness of just how complicated a continuous curve can be. Certainly any curve that contained a segment of the form $C = \{(0, 0)\} \cup \{(x, y) : 0 < x \leq 1, y = x^2 \sin \frac{1}{x}\}$ could not be partitioned as Goursat had envisioned. This led Pringsheim to reject Goursat's theorem and replace it instead with a less ambitious version.

Pringsheim's Theorem. *Suppose f is differentiable at all points inside and on a triangle T. Then $\int_T f(z)\,dz = 0$.*

The proof is a modification of Goursat's argument. That is, let $\int_T f(z)\,dz = I$, and let P denote the perimeter of T. Inserting the lines connecting the midpoints of

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the sides of T, the integral of f around T can be replaced by the integral around the boundaries of the newly created four smaller triangles. Hence, there must be at least one of the four, which we will denote by $T^{(1)}$, whose perimeter is $P/2$ and with $|\int_{T^{(1)}} f(z)\,dz| \geq \frac{I}{4}$. Similarly, connecting the midpoints of the sides of $T^{(1)}$, we can find a triangle $T^{(2)}$ whose perimeter is $P/4$ and with $|\int_{T^{(2)}} f(z)\,dz| \geq \frac{I}{4^2}$. Continuing inductively, we obtain a sequence of triangles $T^{(n)}$ with perimeters equal to $P/2^n$ and with $|\int_{T^{(n)}} f(z)\,dz| \geq \frac{I}{4^n}$.

Let z_0 be the unique point inside all of the triangles $T^{(n)}$. By the differentiability of f at z_0, for any $\varepsilon > 0$, there exists $\delta > 0$ such that whenever $|z - z_0| < \delta$,

$$|f(z) - f(z_0) - (z - z_0)f'(z_0)| \leq \varepsilon|z - z_0|. \tag{4}$$

Choose n sufficiently large so that $T^{(n)}$ lies in a disk of radius δ about z_0 and replace $\int_{T^{(n)}} f(z)\,dz$ by $\int_{T^{(n)}}[f(z) - f(z_0) - (z - z_0)f'(z_0)]\,dz$. Then, according to (4),

$$\left|\int_{T^{(n)}} f(z)\,dz\right| \leq \left|\int_{T^{(n)}} \varepsilon|z - z_0|\,dz\right| \leq \varepsilon\frac{P^2}{4^n}$$

since the perimeter of $T^{(n)}$ is $P/2^n$ and $|z - z_0| \leq P/2^n$ for all points z on $T^{(n)}$. Finally, then,

$$\frac{I}{4^n} \leq \left|\int_{T^{(n)}} f(z)\,dz\right| \leq \varepsilon\frac{P^2}{4^n},$$

and since this holds for all $\varepsilon > 0$, we conclude that $\int_T f(z)\,dz = I = 0$. ■

As an immediate corollary of the triangle theorem, it can be shown that Cauchy's theorem is valid in any convex region D. Pick any z_0 in D, and define $F(z) = \int_{z_0}^{z} f(\zeta)\,d\zeta$, where the path of integration is along the straight line from z_0 to z. By the triangle theorem,

$$\int_{z_0}^{z+h} f(\zeta)\,d\zeta + \int_{z+h}^{z} f(\zeta)\,d\zeta + \int_{z}^{z_0} f(\zeta)\,d\zeta = 0,$$

so that

$$F(z + h) - F(z) = \int_{z}^{z+h} f(\zeta)\,d\zeta,$$

and it follows that $F'(z) = f(z)$ (see, e.g., [**18**, p. 208]). But then $\int_C f(z)\,dz = \int_C F'(z)\,dz = 0$ over any closed curve C.

Since an open disk is obviously convex, it also follows that a function f that is analytic in *any* open set has a local primitive, and the integral of f around piecewise-smooth closed curves is zero, in a sufficiently small disk about any point of the open set.

An essentially identical argument can be made to show that $\int_C f(\zeta)\,d\zeta = 0$ if C is the boundary of a rectangle on which, and inside of which, f is analytic. This also yields the local version of Cauchy's theorem [**14**, pp. 108–109]. In a bit of historical irony, in spite of Pringsheim's strong criticism of Goursat's proof, Pringsheim's triangle theorem (or the equivalent rectangle theorem) have almost universally come to be known as Goursat's theorem. The general closed curve theorem is also often referred to as the Cauchy–Goursat theorem.

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Before we consider the various proofs of the general closed curve theorem, it is interesting to see how a slight variation of Cauchy's original proof falls just short of proving the general result.

6. CAUCHY'S PROOF EXTENDED TO SIMPLY-CONNECTED REGIONS. The following argument is from [**3**, p. 741], although it appeared as early as 1921 in the first edition of [**13**].

Assume that α and β are curves with the same endpoints in a simply-connected region R. Let $\phi(s,t)$ be a twice continuously differentiable map of the unit square $\{(s,t) : 0 \le s,t \le 1\}$ into R with $\phi(0,t) = \alpha(t)$; $\phi(1,t) = \beta(t)$ for all t; and $\phi(s,0) = \alpha(0) = \beta(0)$; $\phi(s,1) = \alpha(1) = \beta(1)$ for all s. Let f be continuously differentiable. Adopting Cauchy's argument, set

$$I(s) = \int_{\phi(s,t)} f(z)\,dz = \int_0^1 f(\phi(s,t))\frac{\partial \phi}{\partial t}\,dt.$$

Then

$$\begin{aligned} I'(s) &= \int_0^1 \Big[f'(\phi(s,t))\frac{\partial \phi}{\partial s}\frac{\partial \phi}{\partial t} + f(\phi(s,t))\frac{\partial^2 \phi}{\partial s \partial t}\Big]\,dt \\ &= \int_0^1 \frac{\partial}{\partial t}\Big[f(\phi(s,t))\frac{\partial \phi}{\partial s}\Big]\,dt = f(\phi(s,t))\frac{\partial \phi}{\partial s}\Big|_{t=0}^{t=1}, \end{aligned}$$

which is zero since $\phi(s,0)$ and $\phi(s,1)$ are constants. (Note that in Cauchy's original proof, $\phi(s,t) = \alpha(t) + s\gamma(t)$ so that $\frac{\partial \phi}{\partial s} = \gamma(t) = 0$ if $t = 0$ or $t = 1$.)

The above argument didn't provide the complete proof of Cauchy's theorem for two reasons. First, while the definition of a simply-connected region assures the existence of a continuous function $\phi(s,t)$ "connecting" the curves α and β (see the definition in Section 7, below), it does not guarantee that ϕ is twice continuously differentiable. Moreover, as we have already seen, the requirement that f' be continuous is actually unnecessary.

As it turned out, however, both of the original approaches suggested by Cauchy and by Goursat can be modified to provide the general proof once the local version of the theorem had been established.

7. COMPLETING THE GENERAL PROOF. The notion of simple-connectedness seems to have assumed its modern form with Poincaré [**16**, p. 84]. (Some important precursors including Riemann's dissertation (1851), perhaps influenced by a short note by Gauss (1840), are discussed in [**17**, pp. 37–38, pp. 59 ff].) There are several equivalent (and common) definitions of simple-connectedness. One definition requires any two curves with the same endpoints in a simply-connected region to be *homotopic*, as defined below.

Definition. Two curves $\alpha, \beta : [0,1] \to D$, with the same endpoints, $\alpha(0) = \beta(0)$ and $\alpha(1) = \beta(1)$, are called homotopic in D (with fixed endpoints), if and only if there exists a continuous map (a so-called homotopy) $H : [0,1] \times [0,1] \to D$ with

(a) $\alpha(t) = H(t,0)$,

(b) $\beta(t) = H(t,1)$,

(c) $\alpha(0) = \beta(0) = H(0,s)$ and $\alpha(1) = \beta(1) = H(1,s)$ for $0 \le s \le 1$.

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The following sequence of lemmas and theorems [**15**, pp. 120 ff], [**14**, pp. 112–117] extend the proof of Cauchy's theorem, showing that the integral of an analytic function in a simply-connected region is identical along any two curves in the region that have the same endpoints. (And the curves in question need only be continuous.)

Lemma. *Let* $\gamma : [0, 1] \to U$ *be a continuous curve. Let* $0 = a_0 < a_1 < a_2 < \cdots < a_n = 1$ *be a partition of* $[0, 1]$ *such that the image* $\gamma([a_i, a_{i+1}])$ *is contained in a disk* D_i *with* $D_i \subset U$. *Suppose* f *is analytic in* U *and let* g_i *be a primitive of* f *on* D_i. *Let* $z_i = \gamma(a_i)$. *Then the sum*

$$\sum_{i=0}^{n-1} [g_i(z_{i+1}) - g_i(z_i)] \tag{5}$$

is independent of the choices of partitions, disks D_i, *and primitives* g_i *on* D_i, *subject to the stated conditions.*

Note that if U is open, then we can find disks $D_i \subset U$, as described in the lemma since γ is continuous and $\gamma([0, 1])$ is compact. Primitives g_i exist in each of the disks D_i as a result of the rectangle/triangle theorem, as noted in Section 5. Both Nevanlinna and Lang note that (5) can be used to extend the definition of $\int_\gamma f$ to any continuous curve, not only those that are piecewise-smooth. It also allows us to show that if two curves with the same endpoints are "close together" (the phrase used by Lang), then the integral of an analytic function is the same along the two curves. Two curves $\gamma, \eta : [0, 1] \to U$ are called "close together" if there exists a partition $0 = a_0 < a_1 < a_2 < \cdots < a_n = 1$, and for each $i = 0, 1, ..., n - 1$, there exists a disk D_i contained in U such that the images of each segment $[a_i, a_{i+1}]$ under the two paths γ, η are contained in D_i.

Cauchy's theorem follows from one final lemma, namely: If $H : [0, 1] \times [0, 1] \to D$ is a homotopy connecting the curves γ and η, then if s_1 and s_2 are sufficiently close, the curves $H(t, s_1)$ and $H(t, s_2)$ are close together so that $\int f$ is the same along those two curves and, by extension, $\int_\gamma f = \int_\eta f$. ■

Much as Lang's argument completed Cauchy's approach, Freitag and Busam [**9**, pp. 233–236] offered an elegant extension of Goursat's approach to show that if f is analytic in a simply-connected region D and γ, η are two curves in D with the same endpoints, then $\int_\gamma f = \int_\eta f$. Their argument, the core of which seems to have first appeared in [**20**], is as follows: Let $\alpha = \gamma - \eta$ represent the closed curve formed by γ followed by the "reverse" of η. Then, if $H : [0, 1] \times [0, 1] \to D$ is a homotopy connecting the curves γ and η, then α is the image of ∂Q, (where $Q = [0, 1] \times [0, 1]$) under the continuous mapping H. By subdividing Q into a union of n^2 congruent smaller squares Q_i , we can express a line integral over α as

$$\int_\alpha f = \sum\nolimits_i \int_{H(\partial Q_i)} f.$$

A standard compactness argument can then be applied to show that, for sufficiently large n, each of these smaller closed curves $H(\partial Q_i)$ is contained in a disk $D_i \subset D$. Finally, as a corollary of the rectangle theorem, since f is analytic in each disk D_i, $\int_{H(\partial Q_i)} f = 0$. By addition, $\int_\alpha f = 0$ and $\int_\gamma f = \int_\eta f$.

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8. ALTERNATIVE DEFINITIONS OF SIMPLE-CONNECTEDNESS. The definition of simple-connectedness given above is one of the two (equivalent) classical ones. The other is that every closed curve in a simply-connected region is homotopic to a constant curve. While these definitions are very general and can be applied in many topological spaces, there are several simpler characterizations of simple-connectedness in the complex plane. For example, a simply-connected region in the plane can be defined as a region that contains the "inside" of any simple closed curve in the region. This definition is based on the Jordan curve theorem that establishes the existence of the two regions known as the inside and the outside of γ, for any simple closed curve γ. Unfortunately, this definition does not lend itself to an easy proof of the closed curve theorem.

A similar, yet more handy, definition can be given directly in terms of complex integrals. It can be shown that, as long as a is not on the closed curve γ,

$$n(\gamma, a) = \frac{1}{2\pi i} \int_\gamma \frac{1}{z-a}\, dz$$

is always integer-valued and determines the number of times that γ "winds around" the point a. Thus, a region R may be defined as simply-connected if, for any closed curve γ contained in R and any point $a \notin R$, $n(\gamma, a) = 0$. This definition, however, does not offer a practical way of determining if a given region is simply-connected, and it is not obviously equivalent to the classical definitions. A more popular approach is found in the influential text by Ahlfors. He defines a region to be simply-connected if its complement is connected in the "extended plane." (The extended plane refers to the topological space obtained by adjoining the point at ∞ to the ordinary complex plane, with sets defined by $|z| > r$ forming the open neighborhoods of ∞. The extended plane is sometimes referred to as the Riemann sphere, to which it is topologically equivalent.) He then shows that any such region satisfies the previous condition. That is, $n(\gamma, a) = \frac{1}{2\pi i} \int_\gamma \frac{1}{z-a}\, dz = 0$ if γ is a closed curve in a simply-connected region and $a \notin R$. This, in turn, leads to a very nice proof of the Cauchy closed curve theorem [**1**, pp. 139–144].

Rudin uses the same definition of simple-connectedness and shows that in a compact subset of a simply-connected region, any function f, analytic in the region, can be uniformly approximated by polynomials [**18**, p. 257]. This leads to a unique and elegant proof of the closed curve theorem since the points of a closed curve γ form a compact set, all polynomials P_n have primitives throughout the complex plane and, if $P_n \to f$ uniformly, then $\int_\gamma f(z)\, dz = \lim_{n\to\infty} \int_\gamma P_n(z)\, dz = 0$.

Rudin [**18**, p. 262] also shows that all of the following statements (as well as several others) regarding a plane region R are equivalent.

(1) Every closed curve in R is homotopic to a constant curve.

(2) The complement of R is connected in the extended plane.

(3) $n(\gamma, a) = \frac{1}{2\pi i} \int_\gamma \frac{1}{z-a}\, dz = 0$ for any closed curve $\gamma \subset R$ and any $a \notin R$.

(4) For every f, analytic in R, there is a function F, analytic in R, and such that $F' = f$.

(5) $\int_\gamma f(z)\, dz = 0$ for any closed curve $\gamma \subset R$ and f analytic in R.

(6) R is homeomorphic to the unit disk.

In particular, then, the alternative definition of a simply-connected plane region, as in (2), is equivalent to the classical (and more general) definition of a simply-connected

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region given by (1). (An example which shows that the analogue of (2) in not equivalent to the classical definitions in general, is the three-dimensional Euclidean space E^3 minus the z-axis: It is not simply-connected, yet its complement in the one-point compactification of E^3 is connected.)

9. NEWMAN'S DEFINITION OF SIMPLE-CONNECTEDNESS. D. J. Newman offered an original definition of a simply-connected region, which also focused on the complement of the region but did not involve the extended plane [**2**, p. 107].

Definition. A plane region R is simply-connected if its complement is "connected within ε to ∞." That is, if for any $z_0 \in \widetilde{R}$ (where $\widetilde{R}$ is the complement of R) and any $\varepsilon > 0$, there is a continuous curve $\eta(t)$, $0 \le t < \infty$, such that

(i) $d(\eta(t), \widetilde{R}) < \varepsilon$ for all $t \ge 0$,
(ii) $\eta(0) = z_0$,
(iii) $\lim_{t\to\infty} \eta(t) = \infty$.

The first condition above says that the curve η always stays within ε of the complement of R, while the other conditions assert that the curve connects z_0 to ∞. The availability of a specific path to work with is what makes this definition so handy. In essence it allows us to use the structure of the real line—a complete linear order—and to talk, e.g., about the last moment of time when the path η remains inside or on a given rectangle. So, using the above definition, it can easily be shown that if a rectangle belongs to a simply-connected region R so do all the points in its interior. Moreover, if we take any closed polygonal path Γ, consisting of horizontal and vertical components contained in R, we can show that its interior is also contained in R. But then, using the rectangle theorem and induction on the number of "levels" of Γ, it follows that the integral of an analytic function around any such closed polygonal path Γ is zero. This allows us to define a primitive F for any function f, analytic in R, as

$$F(z) = \int_{z_1}^{z} f(\zeta)\, d\zeta,$$

where z_1 is a fixed point of R, and the path of integration consists of horizontal and vertical line segments. The Cauchy closed curve theorem follows since, for *any* piecewise-smooth closed curve γ, $\int_\gamma f(z)\, dz = \int_\gamma F'(z)\, dz = F(\gamma(b)) - F(\gamma(a)) = 0$ [**2**, pp. 109–113].

While Newman's definition of simple-connectedness leads to a quick proof of the Cauchy closed curve theorem, it remains to be shown that this definition is equivalent to the classical definition. One-half of the equivalence is easy to establish. If a region R satisfies Newman's condition, then since the Cauchy closed curve theorem is valid in R, according to the "equivalences" listed at the end of the last section, R also satisfies the classic (and other) characterizations of simple-connectedness. But it is not obvious that any region R that satisfies conditions (1) or (2) must satisfy Newman's condition. This is proved below.

10. IF R IS A SIMPLY-CONNECTED REGION, ITS COMPLEMENT, $\widetilde{R}$, IS "CONNECTED WITHIN ε TO ∞." A curve η satisfying (ii) and (iii) in Newman's definition of simple-connectedness is said to "connect z_0 to ∞" or to be a "path from z_0 to ∞." The following example offers some insight into condition (i). Let R be the region in the plane whose complement $\widetilde{R}$ consists of the graph of $y = \sin\frac{\pi}{x}$, $0 < x \le$

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1, together with the part of the y-axis corresponding to $y \geq -1$. Clearly, R is simply-connected, although there is no path in $\widetilde{R}$ from $z_0 = (1, 0)$ to ∞. However, for any $\varepsilon > 0$, there is a path η from z_0 to ∞ satisfying (i). Let $B_\varepsilon(w) = \{z : |z - w| < \varepsilon\}$ and $B_\varepsilon(W) = \cup_{w \in W} B_\varepsilon(w)$, for any point w and any plane set W. Then Newman's definition says that for any $z_0 \in \widetilde{R}$ and any $\varepsilon > 0$ there is a path in $B_\varepsilon(\widetilde{R})$ from z_0 to ∞.

Now let R be any simply-connected plane region. By equivalent condition (2) in Section 8 above, the complement of R in the *extended* plane is connected. It easily follows that every component of $\widetilde{R}$ (the latter being the complement of R in the usual plane) is unbounded: Indeed, if S were such a bounded (and hence compact) component, then the complement of R in the extended plane would have at least two components, namely S and the component containing ∞.

Given any $z_0 \in \widetilde{R}$, let K be the component of $\widetilde{R}$ that contains z_0. We would like to show that, for any $\varepsilon > 0$, there exists a path in $B_\varepsilon(K)$ from z_0 to ∞. Since $B_\varepsilon(K)$ is an unbounded, connected, open set one may wonder if, more generally, there is a path from z_0 to ∞ whenever $z_0 \in V$ and V is any unbounded, connected, open plane set. While the answer to the latter question turns out to be false (as we show below), the idea discussed above, when examined more carefully, is the basis of our proof that any simply-connected plane region satisfies Newman's condition.

Proposition. *There exists an open, connected, unbounded plane set V, whose points are not "connected to ∞" by any path.*

Proof. Let $f(x) = \frac{1}{x}\sin(\frac{\pi}{x})$, and let G be its graph for $0 < x \leq 1$. Then G resembles the well-known graph of $y = \sin\frac{1}{x}$, except that the "amplitude" goes to infinity as we approach the y-axis. Fix any small $p > 0$, and let $V = \{z = (x, y) : 0 < x < 1 + p,\ f(x) - 1 < y < f(x) + 1\}$. Clearly, V is open, and it is obtained as the area swept by G when we slide G up and down by 1 unit (and by adding a bit at the right to make the set open). Let $\mu(t) = (\frac{1}{t+1}, f(\frac{1}{t+1}))$, $t \geq 0$, and $z_0 = (1, 0) = (1, f(1)) = \mu(0)$. Note that μ, as a "path," coincides with G. But formally, μ is not a path from z_0 to ∞ since not only ∞ but every point of the form $(0, y)$ is a limit point of $\mu(t)$ as $t \to \infty$. This is the typical case, and there is no path in V from z_0 to ∞. Any such path would intersect the x-axis near each of the points $z_n = (\frac{1}{n}, 0)$, $n \geq 1$. (The particular choice of z_0 is immaterial; the same would be true for any $z_0 \in V$.) ∎

It is easily seen that the set V in the above example is not of the form $B_\varepsilon(K)$ for any choice of K. We next look more closely at sets that are of this form.

Definition. Let $\varepsilon > 0$. A set U will be called *ε-open* if for every $z \in U$ there is a w_z such that $z \in B_\varepsilon(w_z) \subseteq U$.

Clearly, U is ε-open if and only if it is of the form $B_\varepsilon(W)$ for some set W. If U is ε-open, then it is also ε'-open whenever $0 < \varepsilon' < \varepsilon$. One important property of ε-open sets U is that any bounded sequence in U has a subsequence every two elements of which can be connected by a "short" path in U, as defined below.

Lemma. *Let U be ε-open for some $\varepsilon > 0$, and let $(z_n)_{n=1}^{\infty}$ be a bounded sequence of points in U. Then, given any $\delta > 0$, there is a subsequence σ such that whenever $z_n, z_m \in \sigma$, there is a path γ in U connecting z_n to z_m with* length$(\gamma) < \delta$.

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Proof. Decreasing the value of ε, if necessary, we may assume that $\varepsilon < \frac{\delta}{3}$. For each n, pick w_n with $z_n \in B_\varepsilon(w_n) \subseteq U$. Since the sequence $(w_n)_{n=1}^{\infty}$ is also bounded, by passing to a converging subsequence, we may assume that $|w_n - w_m| < \varepsilon$ for all n, m. Then $w_m \in B_\varepsilon(w_n)$, and hence, the broken path $\gamma = \overline{z_n w_m z_m}$ is in $B_\varepsilon(w_n) \cup B_\varepsilon(w_m) \subseteq U$, and length$(\gamma) < 3\varepsilon < \delta$. ∎

The requirement that U be ε-open for some $\varepsilon > 0$ is essential. Indeed, the sequence $z_n = (\frac{1}{n}, 0)$, $n \geq 1$, of points in V as defined in the proposition given earlier in this section, is bounded, yet it contains no subsequence every two points of which could be connected by a "short" path in V.

The following theorem is our main tool in proving the equivalence of Newman's definition with the usual ones.

Theorem. *Let U be a connected and unbounded plane set that is ε-open for some $\varepsilon > 0$. Then for every $z_0 \in U$ there is a path η in U from z_0 to ∞.*

Proof. Since U is open and connected it is path-connected (see, e.g., [**2**, p. 14]). Since it is also unbounded, there are many paths in U starting at z_0 and ending arbitrarily far away from z_0. We will: (a) take pieces of these paths and put them "one behind the other" (also using the preceding lemma), and (b) make sure these pieces do not come back to z_0 (or to any other finite point) too often so that ∞ is the only limit point of the resulting path η. The details are as follows.

For each $n = 1, 2, \ldots$, the circle C_n centered at z_0 and of radius n must intersect U. Pick $z_n \in C_n \cap U$, and let γ_n be a path from z_0 to z_n in U, with $\gamma_n : [0, 1] \to U$, $\gamma_n(0) = z_0$, $\gamma_n(1) = z_n$.

For each $k = 1, 2, \ldots$ and each $n > k$, the path γ_n must intersect C_k and there must be a "last" point of this intersection, i.e., there exists $t_{k,n} \in (0, 1)$ such that $z_{k,n} := \gamma_n(t_{k,n}) \in C_k$ but $\gamma_n((t_{k,n}, 1]) \cap C_k = \emptyset$ (more precisely, $\gamma_n((t_{k,n}, 1])$ is contained in the unbounded component of $\widetilde{C_k}$). By the preceding lemma, there is an infinite set $S_1 \subseteq \{2, 3, \ldots\}$ such that for all $m, n \in S_1$ there is a path $\omega_{1,m,n}$ from $z_{1,m}$ to $z_{1,n}$ in U with length$(\omega_{1,m,n}) < 1$. There is an infinite set $S_2 \subseteq S_1 \cap \{3, 4, \ldots\}$ such that for all $m, n \in S_2$ there is a path $\omega_{2,m,n}$ from $z_{2,m}$ to $z_{2,n}$ in U with length$(\omega_{2,m,n}) < 1$. By induction, for each $k \geq 2$ there is an infinite set $S_k \subseteq S_{k-1} \cap \{k + 1, k + 2, \ldots\}$ such that for all $m, n \in S_k$ there is a path $\omega_{k,m,n}$ from $z_{k,m}$ to $z_{k,n}$ in U with length$(\omega_{k,m,n}) < 1$.

Now "diagonalize": Let $n_k \in S_k$ for each $k \geq 1$ (with $k < n_k < n_{k+1}$). Construct a path η in U from z_0 to ∞ as follows. Start with $\gamma_{n_1}([0, t_{1,n_1}])$ connecting z_0 to z_{1,n_1}. Follow by ω_{1,n_1,n_2} connecting z_{1,n_1} to z_{1,n_2} by a path of length < 1. Follow by $\gamma_{n_2}([t_{1,n_2}, t_{2,n_2}])$ connecting z_{1,n_2} to z_{2,n_2}. Notice that this path does not reach back into the inside of C_1. Follow by ω_{2,n_2,n_3} connecting z_{2,n_2} to z_{2,n_3}, then by $\gamma_{n_3}([t_{2,n_3}, t_{3,n_3}])$, etc. Let η denote the resulting path. Since each $\omega_{k,n_k,n_{k+1}}$ is of length less than 1 and each $\gamma_{n_{k+1}}([t_{k,n_{k+1}}, t_{k+1,n_{k+1}}])$ misses the inside of C_k, it follows that $\eta(t) \to \infty$ as $t \to \infty$. ∎

Using the main idea described earlier, we can now complete the proof of the equivalence of Newman's definition with the usual ones.

Theorem. *Suppose that R is a simply-connected planar region. Then R is "Newman simply-connected."*

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Proof. Fix $z_0 \in \widetilde{R}$. Let K be the component of z_0 in $\widetilde{R}$. Since R is simply-connected, K must be unbounded. Given $\varepsilon > 0$, the set $U = B_\varepsilon(K)$ is ε-open, unbounded, and connected. Hence, there is a path η in U connecting z_0 to ∞. Clearly, $d(\eta(t), \widetilde{R}) \leq d(\eta(t), K) < \varepsilon$ for all $t \geq 0$, which completes the proof. ∎

11. SUMMARY. The Cauchy closed curve theorem has a long and fascinating history, and although we cannot claim to have told the complete story, we hope that this article conveyed most of the highlights as well as some of the pitfalls in the evolution of the proofs. It is interesting that every cited proof of the general Cauchy closed curve theorem begins with the rectangle theorem of Goursat and the corollary that every function f, which is analytic in a disk, has a primitive F throughout that disk. This is what Newman referred to as the integral theorem for a disk. The integral theorem, in turn, leads to the closed curve theorem. Thus, if C is any closed curve in a disk, $\int_C f = \int_C F' = 0$. (As we noted in Section 5, the rectangle/triangle theorem implies the integral theorem, which then implies the closed curve theorem in any convex region.)

It is also interesting, and a tribute to their great mathematical insight, that the original proofs of Cauchy and Goursat were both able to be updated and expanded to apply to any simply-connected region, even though the definition of a simply-connected region first appeared many years later.

The novelty in Newman's definition is that it allows the general proof to follow the same direct sequence of arguments as the "local" version for functions analytic in a disk. (That is, the closed curve theorem for a rectangle implies the integral theorem for *any* simply-connected region that implies the closed curve theorem for any simply-connected region.)

Our proof of the equivalence of Newman's definition with the usual ones starts with a family of paths (emanating from the same point). We look at the intersection of these paths with bigger and bigger circles, using the compactness of each circle to recursively define a nested sequence of subfamilies such that for each subfamily the intersections of its paths with a corresponding circle cluster to the same point. Using diagonalization, we piece together path segments from each subfamily into a single path that goes to infinity.

D. J. Newman took special pride in his contributions to making mathematics simple. He felt that "beautiful" theorems deserved equally beautiful proofs, and he provided his own elegant proofs for the Bolzano–Weierstrass theorem, the prime number theorem, the minimax theorem, and many others. So it is only fitting that his legacy includes this "simple" proof of the Cauchy closed curve theorem.

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